Maximal Lactate Steady-State Prediction

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The evaluation of blood lactate response to exercise tests provides several physiological- and performance-related indexes, such as lactate turn-points (two turning points may be observed during incremental exercise tests) and the maximal lactate steady-state (MLSS) concept. These indexes have been the subject of extensive research. Several exercise physiologists have addressed questions related to their physiological bases as well as to the effects of exercise test protocols on the measurement of such variables. In addition, these lactate-based indexes have also been employed in numerous human and animal studies as a tool for assessing exercise performance and/or training effects. Most initial questions regarding the biological bases of these indexes and their utilization as an evaluation tool in exercise science were addressed by German scientists. Approximately 30 years ago, the German groups led the research in this field and the results of their ongoing work have contributed notably to generations of exercise scientists to date. Kindermann's group is certainly one of the pioneering groups in this research area. Despite the amount of work done and the reviews published on this topic (lactate-based indexes), the recent review by Faude et al.[1] gathers important data and draws a wide overview of the several lactate threshold (LT) concepts that have emerged in the literature since the 1970s. In their review, Faude et al.[1] evaluated the validity of LT concepts with regard to assessing aerobic endurance capacity or prescribing training intensity (the latter as the agreement between MLSS work-rate [$MLSS_{WR}$] and the elected LTs [LT_{An}]).

The authors accurately indicated that the blood lactate concentration (bLa) at $MLSS_{WR}$ varies considerably among individuals (values from 2 up to 10 mM). Based on these data, they suggest that LT_{An} determination should be performed by individualized approaches rather than using fixed bLa threshold concepts. Later in their review, the authors state that fixed bLa threshold concepts do not take into account interindividual differences and that LT4 may underestimate (particularly in anaerobically trained athletes) or overestimate (in aerobically trained athletes) the $MLSS_{WR}$. Alternatively, the authors present individualized concepts for LT_{An} determination (some of these fixed and individualized concepts have been evaluated in the literature and are cited in their review). Their criticisms of fixed bLa thresholds based on interindividual variations in bLa at $MLSS_{WR}$ seem simplistic and may not take into account all of the data in the literature.[2,3] The classical study by Heck et al.[2] shows, in table II, individual values of $MLSS_{WR}$ and their bLa, as well as the bLa corresponding to $MLSS_{WR}$ as identified during incremental exercise. If a correlation test is performed between MLSS bLa and the incremental exercise bLa corresponding to $MLSS_{WR}$, no relationship is observed between these two variables. This lack of relationship indicates instead that interindividual variations in MLSS bLa may not preclude the validity of fixed bLa thresholds, at least under exercise test conditions similar to those of Heck et al.[2] Nevertheless, the above relationship may indeed exist under slightly different test conditions (lower increment rate) or exercise mode (see Figueira et al.[3] for an overview). Some referenced individualized LT_{AN} approaches[4-8] were previously found to be invalid or their study designs were inappropriate for evaluation of the validity of LT_{AN}. However, this issue is addressed further in another section of their review.

The belief that aerobic training status influences the validity of bLa fixed thresholds has been stated since the 1980s,[4,9,10] but without direct experimental evidence. On the contrary, the only available paper[11] on this matter (i.e. aerobic fitness level vs fixed bLa threshold validity) shows that aerobic fitness level does not affect the validity of fixed threshold (onset of blood lactate accumulation [OBLA] 3.5 mM) to predict $MLSS_{WR}$. It is also known that aerobic training status is not related to bLa at $MLSS_{WR}$,[11,12] which places two of the authors' rationales in conflict with one another. The authors stated that interindividual variations in bLa at $MLSS_{WR}$ may preclude the validity of fixed thresholds; later it is stated that training status would also preclude the validity of fixed thresholds (if the first statement is true, the latter would be in disagreement, since bLa at $MLSS_{WR}$ is not related to training status).

It is recommended, and in fact it is recognized by the authors, that the validity of a given method should be analysed by comparing it with the gold standard measurement (in this case, the direct and individual determination of $MLSS_{WR}$). Faude et al.[1] also appropriately indicated the importance of the Bland and Altman[13] statistical approach in order to assess the validity of a candidate concept. However, in the authors' initial discussion regarding $MLSS_{WR}$ prediction by LT_{AN}, they included many studies whose experimental design or statistical approaches are questionable. Actually, these studies differ from those selected to be in their table V and could have been classified as such (i.e. questionable experimental design). The authors meant that table V includes all of the 11 studies that used the 'recommended procedure' to evaluate the validity of LT_{AN}. Unfortunately, they overlooked some recent studies[3,11,14-17] that would fit the so-called 'recommended procedure' and support the aim of their review. However, two of these studies[16,17] might have appeared in PubMed after the preparation of their review.

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The Authors' Reply

We appreciate the interest of Figueira and colleagues in our recent review[1] of lactate threshold (LT) concepts. A lot of research has

